USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

a. Please fill out the Engineering Change Request Form on the following pages completely:
    a. Detail the names and contact details for each of the ECR contributors
    b. Update the ECR Title
    c. Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
b. For each section/table/figure to be updated:
    a. Detail the section number, starting page and figure/table number to be updated as appropriate.
    b. Detail existing text under "From Text"
    c. Detail changed text under "To Text"
c. Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step a.b
d. Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
    a. This ECR will then be reviewed by the Power Delivery Compliance Working Group.
    b. Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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City: State/Province:

Country: Zip/Postal Code:

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Name: Email:

Company: Mailstop:

Address:

City: State/Province:

Country: Zip/Postal Code:

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: Sink_EPR_Capable_bit_requirement
Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:
As per main specification, EPR_capable bit is not static. Depending on Sink policy it can be set or not during the 1st PD contract. Sink can decide to re issue a new request and set the bit after the first contract.

Benefits as a result of the proposed changes:
Alignment to main specification

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None if DUT properly implemented

An analysis of the hardware implications:
none

An analysis of the software implications:
none

An analysis of the compliance testing implications:
Tester SW shall be updated

An analysis of the Vendor Info File (VIF) implications:
None

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Actual Change Requested

(a). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.EPR.SNK3.2 *EPR Entry Failed – tEnterEPR timeout*

Description: The Tester as a source does not send *EPR_Mode* Enter Succeeded message in the EPR mode sequence and verifies that the UUT sends *Soft Reset* Message.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
    b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. The check fails if the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure and the test stops here [TEST.PD.EPR.SNK3.2#1]

To:

TEST.PD.EPR.SNK3.2 *EPR Entry Failed – tEnterEPR timeout*

Description: The Tester as a source does not send *EPR_Mode* Enter Succeeded message in the EPR mode sequence and verifies that the UUT sends *Soft Reset* Message.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. ~~The check fails if~~ If the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure

a) the Tester waits for an other Request Message for Time_To_EPR_Ready from VIF and checks that “EPR mode capable” field is set

b) If no new Request is received the Tester fails the check and the test stops here [TEST.PD.EPR.SNK3.2#1]

(b). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.EPR.SNK3.3 *EPR Entry Failed by EPR_Mode (Enter failed) Message*

Description: :As a Source, the Tester verifies that the UUT responds with a *Soft Reset* Message to the *EPR_Mode* Failed message.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
   a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
   b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. The check fails if the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure and the test stops here [TEST.PD.EPR.SNK3.3#1]

To:

TEST.PD.EPR.SNK3.3 *EPR Entry Failed by EPR_Mode (Enter failed) Message*

Description: :As a Source, the Tester verifies that the UUT responds with a *Soft Reset* Message to the *EPR_Mode* Failed message.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. ~~The check fails~~ If the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure

a) the Tester waits for an other Request Message for Time_To_EPR_Ready from VIF and checks that “EPR mode capable” field is set

b) If no new Request is received the Tester fails the check and the test stops here [TEST.PD.EPR.SNK3.3#1]

(c). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.EPR.SNK3.4 *EPR Entry Failed due to tFirstSourceCap timeout*

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for Source Capability timeout by Tester

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓
Rev3UnchkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. The check fails if the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure and the test stops here [TEST.PD.EPR.SNK3.4#1]

To:

TEST.PD.EPR.SNK3.4 *EPR Entry Failed due to tFirstSourceCap timeout*

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for Source Capability timeout by Tester

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓
Rev3UnchkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
    a) The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
    b) The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. ~~The check fails~~ If the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure
    a) the Tester waits for an other Request Message for Time_To_EPR_Ready from VIF and checks that “EPR mode capable” field is set
    b) If no new Request is received the Tester fails the check and the test stops here [TEST.PD.EPR.SNK3.4#1]

(d). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.EPR.SNK3.5 *EPR Exit due to incorrect EPR Source Cap*

Description: The Tester as a source verifies the EPR Exit due to the incorrect EPR Source Capability

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

   a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

   b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. The check fails if the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure and the test stops here [TEST.PD.EPR.SNK3.5#1]

To:

TEST.PD.EPR.SNK3.5 *EPR Exit due to incorrect EPR Source Cap*

Description: The Tester as a source verifies the EPR Exit due to the incorrect EPR Source Capability

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

   a) The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

   b) The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. ~~The check fails~~ If the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure

   a) the Tester waits for an other Request Message for Time_To_EPR_Ready from VIF and checks that "EPR mode capable" field is set

b) If no new Request is received the Tester fails the check and the test stops here [TEST.PD.EPR.SNK3.5#1]

(e). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.EPR.SNK3.6 EPR Exit due to EPR_Mode Exit Message

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for *EPR_Mode* (Exit) Message (In the EPR PD Negotiation)

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
    1. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
    2. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. The check fails if the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure and the test stops here [TEST.PD.EPR.SNK3.6#1]

To:

TEST.PD.EPR.SNK3.6 EPR Exit due to EPR_Mode Exit Message

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for *EPR_Mode* (Exit) Message (In the EPR PD Negotiation)

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

   a) The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

   b) The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. ~~The check fails~~ If the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure

   a) the Tester waits for an other Request Message for Time_To_EPR_Ready from VIF and checks that “EPR mode capable” field is set

   b) If no new Request is received the Tester fails the check and the test stops here [TEST.PD.EPR.SNK3.6#1]

(f). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.EPR.SNK3.8 *EPR Exit due to Source Cap*

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for the Source Capability message after the EPR PD negotiation.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter EPR_Supported_As_Snk is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

   a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. The check fails if the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure and the test stops here [TEST.PD.EPR.SNK3.8#1]

To:

TEST.PD.EPR.SNK3.8 *EPR Exit due to Source Cap*

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for the Source Capability message after the EPR PD negotiation.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter EPR_Supported_As_Snk is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

1. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

2. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. ~~The check fails~~ If the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure

a) the Tester waits for an other Request Message for Time_To_EPR_Ready from VIF and checks that "EPR mode capable" field is set

b) If no new Request is received the Tester fails the ceheck and the test stops here [TEST.PD.EPR.SNK3.8#1]

(g). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.EPR.SNK3.9 EPR Entry failed due to Source Cap

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for the *Source_Capabilities* Message during the EPR PD negotiation.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
    b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. The check fails if the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure and the test stops here [TEST.PD.EPR.SNK3.9#1]

To:

TEST.PD.EPR.SNK3.9 EPR Entry failed due to Source Cap

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for the *Source_Capabilities* Message during the EPR PD negotiation.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

   a) The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

   b) The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. ~~The check fails~~ If the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure

   a) the Tester waits for an other Request Message for Time_To_EPR_Ready from VIF and checks that "EPR mode capable" field is set

   b) If no new Request is received the Tester fails the check and the test stops here [TEST.PD.EPR.SNK3.9#1]

(h). Section x.x.x, Page x, Figure/Table x-x

FROM:

TEST.PD.EPR.SNK3.10 *EPR Exit fail due to SinkWaitCapTimer timeout*

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for *SinkWaitCapTimer* timeout by Tester during the EPR Exit process.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter EPR_Supported_As_Snk is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓
Rev3UnchkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
   a) The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
   b) The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. The check fails if the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure and the test stops here. [TEST.PD.EPR.SNK3.10#1]

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To:

TEST.PD.EPR.SNK3.10 *EPR Exit fail due to SinkWaitCapTimer timeout*

Description: The Tester as a source verifies the UUT responds with *Hard Reset* for *SinkWaitCapTimer* timeout by Tester during the EPR Exit process.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter EPR_Supported_As_Snk is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓
Rev3UnchkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
   a) The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
   b) The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. ~~The check fails~~ If the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure

   a) the Tester waits for an other Request Message for Time_To_EPR_Ready from VIF and checks that "EPR mode capable" field is set

   b) If no new Request is received the Tester fails the check and the test stops here. [TEST.PD.EPR.SNK3.10#1]